

Unit 5

CHEMICAL ENERGETICS

30 items · 77 marks

Syllabus sections in this unit

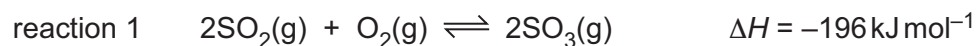
5.1 Enthalpy change, ΔH

5.2 Hess's law

Unit 5: Chemical energetics**1. M/J 26/P23/Q3(a)**

Context

- 3** The conditions used for the conversion of sulfur dioxide to sulfur trioxide in the Contact process are carefully controlled.



- (a)** Complete Figure 3.1 to show the reaction pathway diagram for reaction 1.

Draw arrows to represent the enthalpy change, ΔH , and activation energy, E_A .

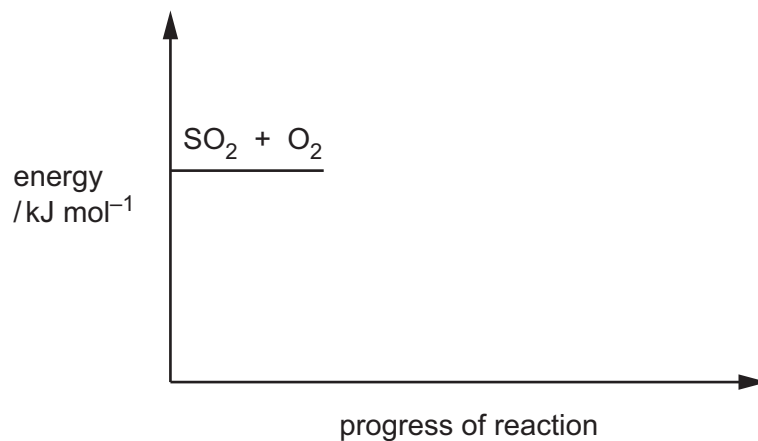


Figure 3.1

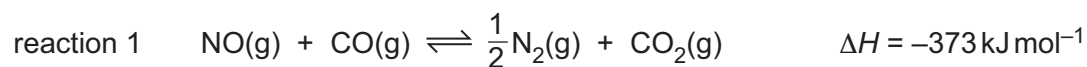
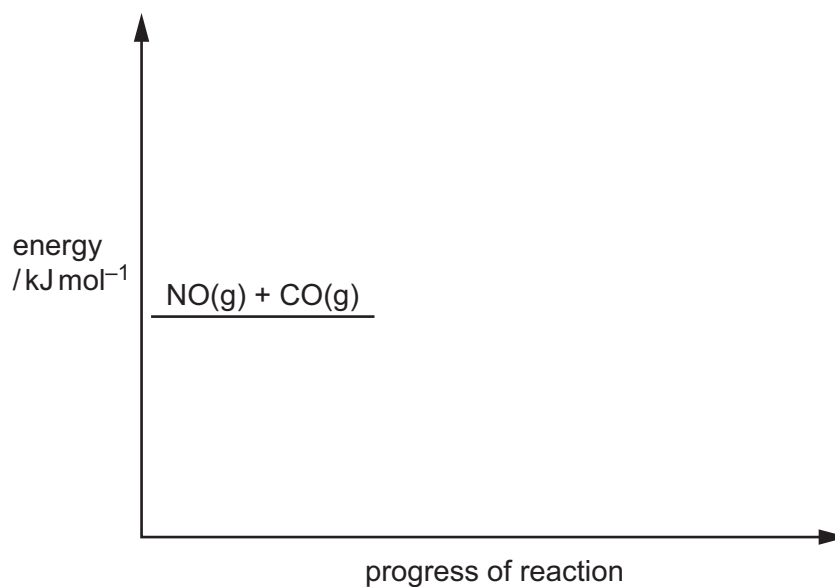
[2]

2. MAR 26/P22/Q2(d)(i)

Context

2 Nitrogen monoxide, NO, is an atmospheric pollutant.

Context: part (d) (introduction)

(d) Reaction 1 occurs during the catalytic removal of NO from car exhaust gases.The activation energy, E_a , of this reaction is 570 kJ mol^{-1} .**(i)** Complete Figure 2.3 to sketch the energy profile for reaction 1.Show the enthalpy change, ΔH , and the activation energy, E_a .**Figure 2.3**

[2]

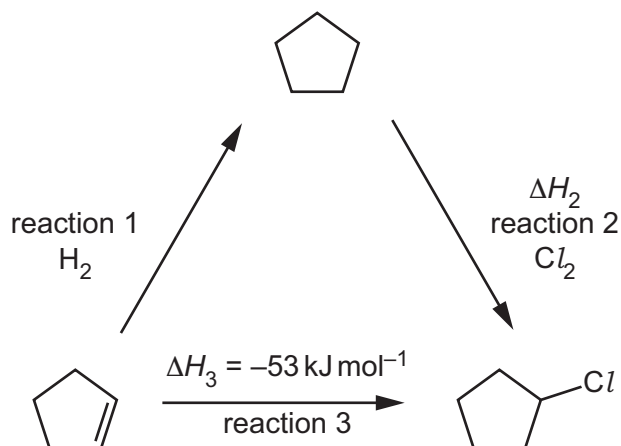
3. O/N 25/P21/Q3(d)(ii)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

Context: part (d) (introduction)

- (d)** Fig. 3.2 shows a reaction cycle involving cyclopentane, cyclopentene and $\text{C}_5\text{H}_9\text{Cl}$.

**Fig. 3.2**

- (ii)** Use the data in Fig. 3.2 and in Table 3.1 to calculate the enthalpy change of reaction 2, ΔH_2 .

Table 3.1

compound	enthalpy change of combustion, $\Delta H_c / \text{kJ mol}^{-1}$
	-3292
	-3115
H_2	-286

$\Delta H_2 =$ kJ mol⁻¹ [2]

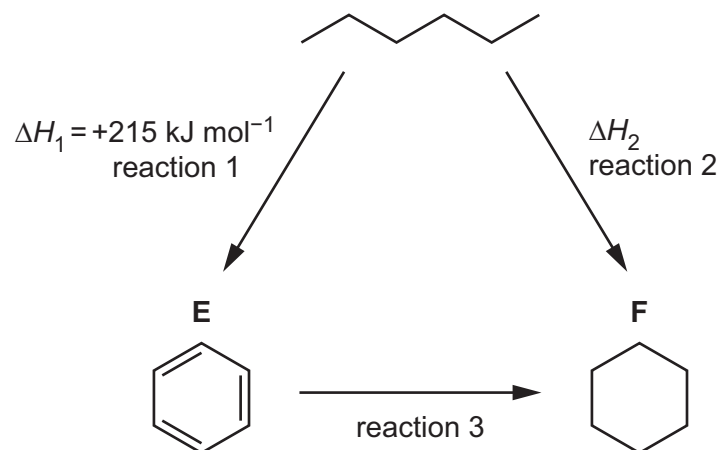
4. O/N 25/P22/Q3(e)(ii)

Context

- 3** The alkanes are a homologous series of organic molecules. Alkanes are generally unreactive and are commonly used as fuels.

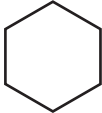
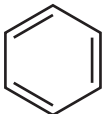
Context: part (e) (introduction)

- (e)** Hexane can be converted into compounds **E** and **F** at high temperature and pressure. Fig. 3.2 shows the reaction scheme involving hexane, **E** and **F**.

**Fig. 3.2**

- (ii)** Use the data in Fig. 3.2 and in Table 3.1 to calculate the enthalpy change of reaction 2, ΔH_2 .

Table 3.1

compound	enthalpy change of formation, $\Delta H_f / \text{kJ mol}^{-1}$
	-156
	+48

 $\Delta H_2 =$ kJ mol^{-1} [2]

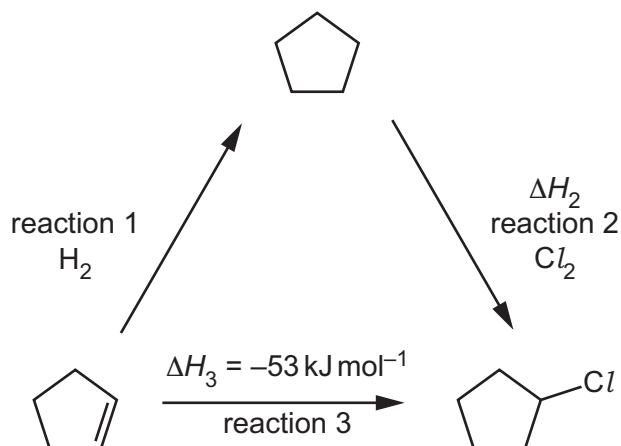
5. O/N 25/P23/Q3(d)(ii)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

Context: part (d) (introduction)

- (d)** Fig. 3.2 shows a reaction cycle involving cyclopentane, cyclopentene and $\text{C}_5\text{H}_9\text{Cl}$.

**Fig. 3.2**

- (ii)** Use the data in Fig. 3.2 and in Table 3.1 to calculate the enthalpy change of reaction 2, ΔH_2 .

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compound	enthalpy change of combustion, $\Delta H_c / \text{kJ mol}^{-1}$
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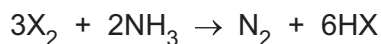
$\Delta H_2 =$ kJ mol⁻¹ [2]

6. O/N 25/P24/Q1(b)(iii)-(iv)

Context

1 The Group 17 elements are oxidising agents.

Context: part (b) (introduction)

(b) Cl_2 and Br_2 can each react with NH_3 to give N_2 and a hydrogen halide, HX . $\text{X} = \text{Cl or Br}$

The relative bond strengths of X-X and H-X determine the difference in enthalpy change of the two reactions.

(iii) Define enthalpy change of formation, ΔH_f .

[2]

(iv) Table 1.1 gives data relevant to the reaction of $\text{Cl}_2(\text{g})$ with $\text{NH}_3(\text{g})$.**Table 1.1**

compound	enthalpy change of formation, $\Delta H_f / \text{kJ mol}^{-1}$
$\text{NH}_3(\text{g})$	-46
$\text{HCl}(\text{g})$	-92

Use the data in Table 1.1 to calculate the enthalpy change of the reaction of $\text{Cl}_2(\text{g})$ with $\text{NH}_3(\text{g})$.

enthalpy change of reaction = kJ mol^{-1} [2]

7. M/J 25/P22/Q3(c)**(c)** Hydrocarbon **Y** contains two $\text{C}=\text{C}$ groups. There are no other functional groups present.**Y** reacts with an excess of $\text{H}_2(\text{g})$ to produce hexane, C_6H_{14} .**Table 3.1**

bond	energy / kJ mol^{-1}
C-C	350
$\text{C}=\text{C}$	610
$\text{C}\equiv\text{C}$	840
H-H	436
C-H	410

Use Table 3.1 to calculate the enthalpy change per mole of C_6H_{14} produced in this reaction.

enthalpy change = kJ mol^{-1} [2]

8. M/J 24/P21/Q4(a)**4** **(a)** Define enthalpy change of formation.

[2]

9. M/J 24/P21/Q4(b)

(b) Iron is made when iron(III) oxide is heated with carbon monoxide, as shown by reaction 2.

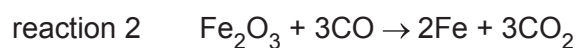


Table 4.1 shows enthalpy change of formation data measured at 298 K and 101 kPa.

Table 4.1

substance	equation	value for $\Delta H_f^\ominus / \text{kJ mol}^{-1}$
Fe_2O_3		–824.2
CO		–110.5
CO_2	$\text{C(s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$	–393.5

(i) Complete Table 4.1 by adding equations with relevant state symbols to represent:

- standard enthalpy change of formation for Fe_2O_3
- standard enthalpy change of formation for CO.

[2]

(ii) Use the data in Table 4.1 to calculate the enthalpy change of reaction, ΔH_r , in kJ mol^{-1} , for reaction 2.

Show your working.

$\Delta H_r =$ kJ mol^{-1} [2]

10. M/J 24/P22/Q2(a)

Context

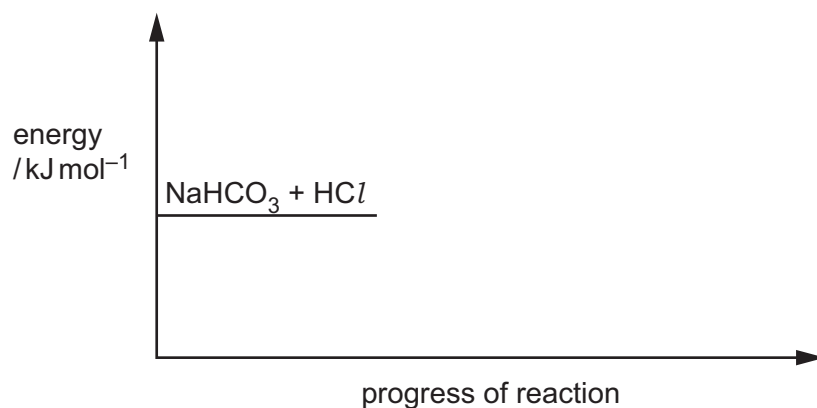
- 2** Separate samples of Na_2CO_3 and NaHCO_3 react with $\text{HCl}(\text{aq})$ to produce the same products, as shown in Table 2.1.

Table 2.1

reaction	equation	$\Delta H/\text{kJ mol}^{-1}$
1	$\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$	ΔH_1
2	$\text{NaHCO}_3 + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$	$\Delta H_2 = +27.2$

- (a)** Complete the reaction pathway diagram in Fig. 2.1 for reaction 2.

Label the diagram to show the enthalpy change, ΔH_2 , and the activation energy, E_A .

**Fig. 2.1**

[2]

11. M/J 24/P22/Q2(b)

Context

- 2** Separate samples of Na_2CO_3 and NaHCO_3 react with $\text{HCl}(\text{aq})$ to produce the same products, as shown in Table 2.1.

Table 2.1

reaction	equation	$\Delta H/\text{kJ mol}^{-1}$
1	$\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$	ΔH_1
2	$\text{NaHCO}_3 + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$	$\Delta H_2 = +27.2$

- (b)** The value for ΔH_1 is determined by experiment using the following method.

- 50.0 cm^3 of 2.00 mol dm^{-3} $\text{HCl}(\text{aq})$ is added to a polystyrene cup.
- The initial temperature of the acid is recorded as 19.6°C .
- 0.0400 mol of Na_2CO_3 is added and the mixture is stirred.
- All the solid Na_2CO_3 disappears and a colourless solution is produced.

The maximum temperature recorded during the reaction is 26.2°C .

- (i)** Describe **one** other observation that shows the reaction is complete.

[1]

5

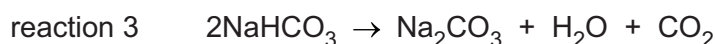
- (ii)** Calculate the value of ΔH_1 in kJ mol^{-1} .

Assume the specific heat capacity of the reaction mixture is the same as for water and no heat is lost to the surroundings.

Show your working.

$$\Delta H_1 = \quad \text{kJ mol}^{-1} \quad [3]$$

- (iii)** Thermal decomposition occurs when NaHCO_3 is heated.



Calculate the enthalpy change for reaction 3, ΔH_r , using the data in Table 2.1 and the value of ΔH_1 calculated in **(b)(ii)**.

(If you were unable to calculate a value for ΔH_1 in **(b)(ii)**, assume the enthalpy change is -38.4 kJ mol^{-1} . This is **not** the correct value.)

$$\Delta H_r = \quad \text{kJ mol}^{-1} \quad [2]$$

12. M/J 24/P23/Q3(a)

- 3** **(a)** Write an equation to show the reaction for the standard enthalpy change of formation of H_2O . Include state symbols.

[2]

13. M/J 24/P23/Q3(b)

(b) Water is one of the products in the reaction of B_2O_3 and NH_3 , as shown in reaction 2.

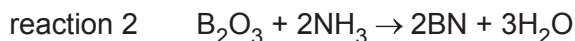


Table 3.1 shows information about the standard enthalpy change of formation, $\Delta H_f^\ominus$, of some substances.

Table 3.1

substance	$\Delta H_f^\ominus / \text{kJ mol}^{-1}$
B_2O_3	-1264
NH_3	-46
BN	-134
H_2O	-286

Calculate the enthalpy change, ΔH , for reaction 2 using the data from Table 3.1.

$$\Delta H = \quad \quad \quad \text{kJ mol}^{-1} \quad [2]$$

14. MAR 24/P22/Q3(c)(i)

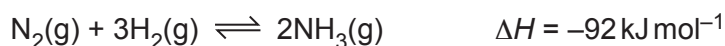
Context

3 Nitrogen, N_2 , is generally an unreactive molecule but it does react under certain conditions.

Context: part (c) (introduction)

(c) The Haber process involves the reaction of N_2 and H_2 to form ammonia, NH_3 .

A catalyst is used, which allows the process to be carried out at a lower temperature and pressure.



(i) Use the information in (c) to complete Table 3.1.

Table 3.1

compound	enthalpy change of formation, $\Delta H_f^\ominus / \text{kJ mol}^{-1}$
N_2	
H_2	
NH_3	

[2]

15. O/N 23/P21/Q1(e)

Context

- 1** The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

- (e) POCl₃(g) forms when PCl₃(g) reacts with O₂(g).

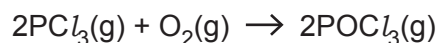


Table 1.3 gives some relevant data.

Table 1.3

process	value /kJ mol ⁻¹
enthalpy change of formation of PCl ₃ (g)	-289
enthalpy change of formation of POCl ₃ (g)	-592
O ₂ (g) → 2O(g)	+496

- (i) Define enthalpy change of formation, ΔH_f° .

[2]

- (ii) Calculate the bond energy of P=O in POCl₃ using the data in Table 1.3.

Show your working.

bond energy of P=O =

kJ mol⁻¹

[2]

16. O/N 23/P22/Q3(b)(i)-(ii)

Context

3 Phosphoric(V) acid, H_3PO_4 , is used in both inorganic and organic reactions.

Context: part (b) (introduction)

(b) H_3PO_4 is also formed in the process shown in reaction 1.

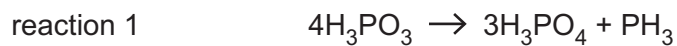


Table 3.1 shows some relevant thermodynamic data.

Table 3.1

compound	enthalpy change of formation, $\Delta H_f / \text{kJ mol}^{-1}$
H_3PO_3	-972
H_3PO_4	-1281
PH_3	+9

(i) Define enthalpy change of formation.

[2]

(ii) Use the data in Table 3.1 to calculate the enthalpy change, ΔH_r , of reaction 1.

$\Delta H_r =$ kJ mol^{-1}
[2]

17. O/N 23/P23/Q1(e)

Context

- 1** The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy / kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

- (e) POC_l₃(g) forms when PC_l₃(g) reacts with O₂(g).

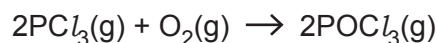


Table 1.3 gives some relevant data.

Table 1.3

process	value / kJ mol ⁻¹
enthalpy change of formation of PC _l ₃ (g)	-289
enthalpy change of formation of POC _l ₃ (g)	-592
O ₂ (g) → 2O(g)	+496

- (i) Define enthalpy change of formation, ΔH_f.

[2]

- (ii) Calculate the bond energy of P=O in POC_l₃ using the data in Table 1.3.

Show your working.

bond energy of P=O =

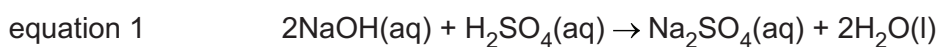
kJ mol⁻¹

[2]

18. M/J 23/P21/Q2(a)

Context

- 2** A neutralisation reaction occurs when NaOH(aq) is added to H₂SO₄(aq).



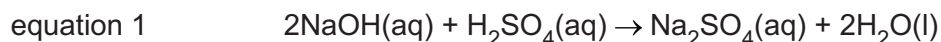
- (a) Define enthalpy change of neutralisation, ΔH_{neut}.

[2]

19. M/J 23/P21/Q2(b)(ii)

Context

2 A neutralisation reaction occurs when NaOH(aq) is added to H₂SO₄(aq).



Context: part (b) (introduction)

(b) An experiment is carried out to calculate ΔH_{neut} for the reaction between NaOH(aq) and H₂SO₄(aq).

100 cm³ of 1.00 mol dm⁻³ NaOH(aq) is added to 75 cm³ of 1.00 mol dm⁻³ H₂SO₄(aq) in a polystyrene cup and stirred. Results from the experiment are shown in Table 2.1.

Table 2.1

initial temperature of NaOH(aq)/°C	20.0
initial temperature of H ₂ SO ₄ (aq)/°C	20.0
maximum temperature of mixture/°C	27.8

(ii) Calculate ΔH_{neut} using the results in Table 2.1. Include units in your answer.

Assume that:

- the specific heat capacity of the final solution is 4.18 J g⁻¹ K⁻¹
- 1.00 cm³ of the final solution has a mass of 1.00 g
- there is no heat loss to the surroundings
- full dissociation of H₂SO₄(aq) occurs
- the experiment takes place at constant pressure.

Show your working.

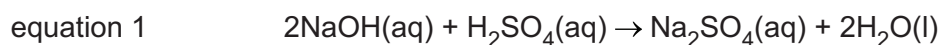
$$\Delta H_{\text{neut}} = \quad \text{units}$$

[3]

20. M/J 23/P21/Q2(c)(ii)

Context

2 A neutralisation reaction occurs when NaOH(aq) is added to H₂SO₄(aq).



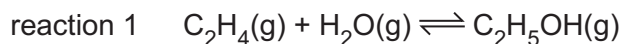
(ii) Suggest why the enthalpy change of neutralisation cannot be determined using the addition of dilute sulfuric acid to aqueous barium hydroxide.

[1]

21. M/J 23/P22/Q4(a)

Context

- 4** In industry, ethanol is made by reacting ethene with steam in the presence of H_3PO_4 .



- (a) Use the bond energy values in Table 4.1 to calculate the enthalpy change, ΔH_r , for reaction 1.

Table 4.1

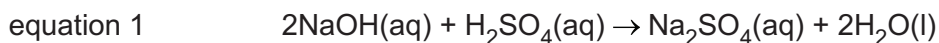
bond	bond energy / kJ mol^{-1}
C–C	350
C=C	610
C≡C	840
C–H	410
C–O	360
C=O	740
O–H	460

$$\Delta H_r = \quad \text{kJ mol}^{-1} \quad [2]$$

22. M/J 23/P23/Q3(a)

Context

- 3** A neutralisation reaction occurs when $\text{NaOH}(\text{aq})$ is added to $\text{H}_2\text{SO}_4(\text{aq})$.

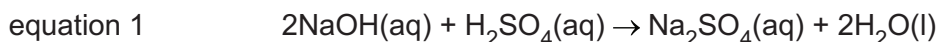


- (a) Define enthalpy change of neutralisation, ΔH_{neut} .

[2]

23. M/J 23/P23/Q3(b)(ii)

Context

3 A neutralisation reaction occurs when NaOH(aq) is added to H₂SO₄(aq).

Context: part (b) (introduction)

(b) In an experiment, 50.0 cm³ of 2.00 mol dm⁻³ NaOH(aq) is added to 60.0 cm³ of 1.00 mol dm⁻³ H₂SO₄(aq) in a polystyrene cup and stirred. Both solutions have a temperature of 21.4 °C before mixing. The maximum temperature of the mixture is measured.

(ii) Calculate the theoretical maximum temperature of the mixture in this experiment.

Assume that:

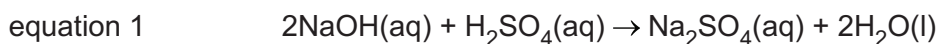
- enthalpy change of neutralisation, ΔH_{neut} , of NaOH(aq) and H₂SO₄(aq) is -57.1 kJ mol⁻¹
- full dissociation of H₂SO₄(aq) occurs
- the specific heat capacity of the final solution is 4.18 J g⁻¹ K⁻¹
- 1.00 cm³ of the final solution has a mass of 1.00 g
- there is no heat loss to the surroundings
- the experiment takes place at constant pressure.

Show your working.

theoretical maximum temperature = °C [3]

24. M/J 23/P23/Q3(c)(ii)

Context

3 A neutralisation reaction occurs when NaOH(aq) is added to H₂SO₄(aq).

Context: part (c) (introduction)

(c) The enthalpy change of neutralisation of CH₃COOH(aq) and NaOH(aq) is -55.2 kJ mol⁻¹.

(ii) Values for the enthalpy change of neutralisation, ΔH_{neut} , are shown in Table 3.1.

Table 3.1

reagents	$\Delta H_{\text{neut}}/\text{kJ mol}^{-1}$
NaOH + HCl	-57.1
NaOH + CH ₃ COOH	-55.2

Suggest why the value for ΔH_{neut} of the weak acid, CH₃COOH, reacting with NaOH is different to the value obtained using the strong acid, HCl. Assume that the values are determined under the same conditions.

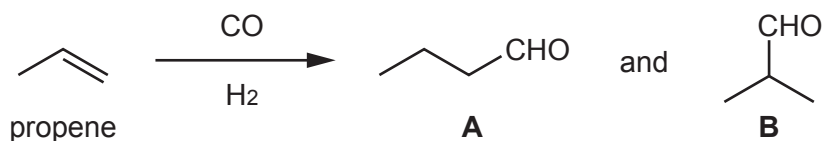
[1]

25. MAR 23/P22/Q3(d)(iii)

Context

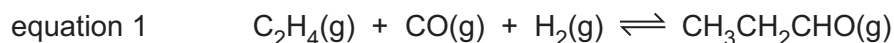
- 3** Alkenes undergo an addition reaction with a 1:1 mixture of CO and H₂ to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H₂.

**Fig. 3.1**

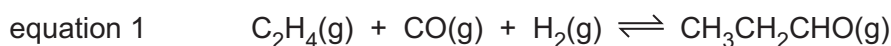
Context: part (d) (introduction)

- (d)** The reaction of ethene, C₂H₄, with a 1:1 mixture of CO and H₂ is shown in equation 1.



At atmospheric pressure a cobalt-based catalyst is used in this reaction.

- (iii)** Use the information in Table 3.2 to calculate the enthalpy change, ΔH_r , of the reaction in equation 1.

**Table 3.2**

compound	enthalpy change of formation, $\Delta H_f^\circ/\text{kJ mol}^{-1}$
C ₂ H ₄ (g)	+52
CO(g)	−111
CH ₃ CH ₂ CHO(g)	−187

$\Delta H_r =$ kJ mol^{−1} [2]

26. M/J 22/P21/Q1(f)(ii)

Context

- 1** Calcium, magnesium and radium are Group 2 elements. Radium follows the same trends as the other members of Group 2.

Context: part (f) (introduction)

- (f)** Magnesium, Mg, burns in oxygen, O₂.
The activation energy, E_a , for this reaction is +148 kJ mol⁻¹.

(ii) On Fig. 1.1:

- sketch a reaction pathway diagram for the reaction that occurs when Mg burns in O₂
- label the diagram to show the enthalpy change, ΔH , and the activation energy, E_a , for the reaction.

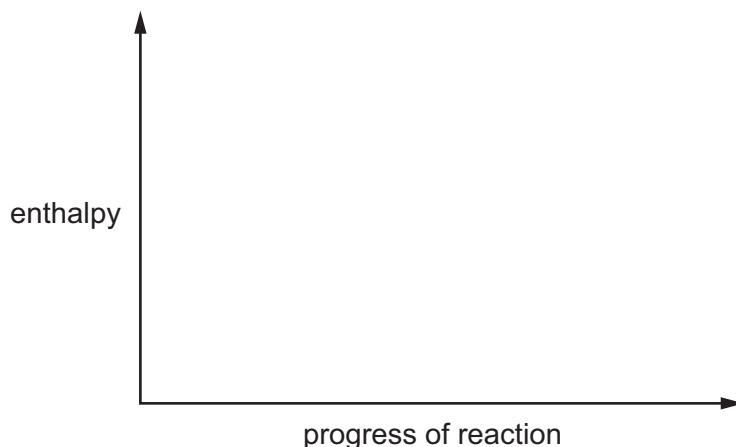


Fig. 1.1

[3]

27. M/J 22/P22/Q1(b)(iii)-(iv)

Context: part (b) (introduction)

- (b)** When magnesium is heated in air, magnesium oxide, MgO, is the major product. Smaller amounts of magnesium nitride, Mg₃N₂, are also made.

(iii) Define enthalpy change of formation.

[2]

- (iv)** When 3.645 g of Mg(s) burns in excess N₂(g) to form Mg₃N₂(s), 23.05 kJ of energy is released.

Calculate the enthalpy change of formation, ΔH_f , of Mg₃N₂. Show your working.

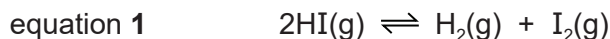
$$\Delta H_f (\text{Mg}_3\text{N}_2) =$$

[3]

28. M/J 22/P22/Q3(b)

Context: part (a) (introduction)

- 3 (a)** 0.025 mol of HI(g) is added to a closed vessel and left to reach dynamic equilibrium. The total pressure of the vessel is 100 kPa.



- (b)** Use equation 1 and the bond energy values in Table 3.1 to calculate the change in enthalpy, ΔH , for the thermal decomposition of 1 mole of HI(g). Show your working.

Table 3.1

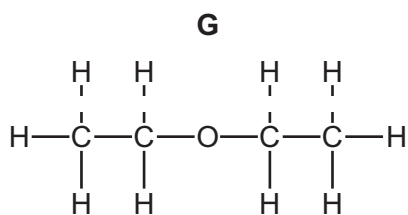
bond	bond energy / kJ mol^{-1}
H–H	436
I–I	151
H–I	299

$$\Delta H = \quad \text{kJ mol}^{-1} \quad [2]$$

29. M/J 22/P23/Q3(a)(ii)

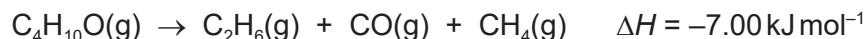
Context

- 3 G** belongs to a group of compounds called ethers.

**Fig. 3.1**

Context: part (a) (introduction)

- (a)** When **G**, $\text{C}_4\text{H}_{10}\text{O}$, is heated, thermal decomposition occurs.



- (ii)** Calculate the bond energy of $\text{C}\equiv\text{O}$ using the bond energy values in Table 3.1 and the enthalpy change, ΔH , for the thermal decomposition of **G**. Show your working.

Table 3.1

bond	bond energy / kJ mol^{-1}
C–C	350
C–O (in G)	360
C–H	410

$$\text{bond energy (C}\equiv\text{O)} = \quad \text{kJ mol}^{-1} \quad [2]$$

30. MAR 22/P22/Q2(c)(iii)

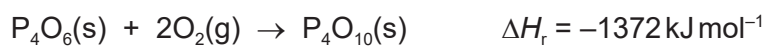
Context

2 Some oxides of elements in Period 3 are shown.

Context: part (c) (introduction)

(c) P_4O_6 is a white solid that has a melting point of 24°C . Solid P_4O_6 reacts with water to form H_3PO_3 .

(iii) When $\text{P}_4\text{O}_6(\text{s})$ is heated with oxygen it forms $\text{P}_4\text{O}_{10}(\text{s})$.



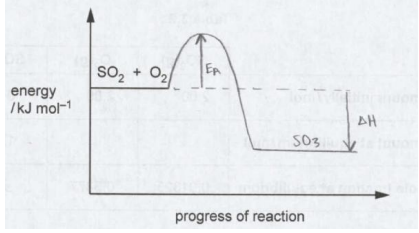
The enthalpy change of formation, ΔH_f , of $\text{P}_4\text{O}_{10}(\text{s})$ is $-3012 \text{ kJ mol}^{-1}$.

Calculate the enthalpy change of formation, ΔH_f , of $\text{P}_4\text{O}_6(\text{s})$.

ΔH_f of $\text{P}_4\text{O}_6(\text{s}) =$ kJ mol^{-1} [1]

Answers Section

1. M/J 26/P23/Q3(a)

3(a)	 • correct shape profile • arrow label ΔH from reactant line to product line • arrow label E_a (from reactant line to maximum on hump)	•✓✓	2 max 1 endothermic profile with correct labels & arrows for E_a and / or ΔH ΔH single headed arrows used in correct direction only. A E_a shown as a double headed arrow BOD vertical line A half-headed arrows A 'square' humps I missing products / incorrect labels on product line I '-' sign on $-\Delta H$ BOD poorly represented lines / arrows if approximate position and length (and direction) is correct
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2. MAR 26/P22/Q2(d)(i)

2(d)(i)	• reactants higher than labelled products AND hump in the middle • reactants to top of hump = E_a OR (+)570 (kJ mol⁻¹) • reactants to products = ΔH OR (-)373 (kJ mol⁻¹)	2
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3. O/N 25/P21/Q3(d)(ii)

3(d)(ii)	M1 (reaction 1): $-3115 - 286 + 3292 = -109$ (kJ mol ⁻¹) M2 $\Delta H_2 = -53 - (-109) = (+)56$ (kJ mol ⁻¹)	2
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4. O/N 25/P22/Q3(e)(ii)

3(e)(ii)	M1 $+215 + (-156) - (+48) =$ M2 $+11$ kJ mol ⁻¹	2
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5. O/N 25/P23/Q3(d)(ii)

3(d)(ii)	M1 (reaction 1): $-3115 - 286 + 3292 = -109$ (kJ mol ⁻¹) M2 $\Delta H_2 = -53 - (-109) = (+)56$ (kJ mol ⁻¹)	2
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6. O/N 25/P24/Q1(b)(iii)-(iv)

1(b)(iii)	M1 (enthalpy change / energy change) when one mole of a compound / substance is formed M2 from its (constituent) elements in their standard state(s)	2
1(b)(iv)	M1 $(6 \times -92) - (2 \times -46) =$ M2 -460 (kJ mol ⁻¹)	2

7. M/J 25/P22/Q3(c)

3(c)	method 1 – breaking all bonds in molecules $(2 \times 610) + (2 \times 436) + (10 \times 410) + (3 \times 350) - (14 \times 410) - (5 \times 350)$ - energy involved when all bonds are broken = (7242) - energy involved when all bonds are made = (7490) - energy to break bonds – energy to make bonds $= (7242 - 7490) = -248$ method 2 – ignore any bonds that do not change from reactants to products $(2 \times 610) + (2 \times 436) - (4 \times 410) - (2 \times 350)$ - energy involved when all bonds are broken = (2092) - energy involved when all bonds are made = (2340) - energy to break bonds – energy to make bonds $(2092 - 2340) = -248$	2
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8. M/J 24/P21/Q4(a)

4(a)	M1 (enthalpy change / energy change) when one mole of a compound / substance is formed M2 from its elements in their standard states	2
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9. M/J 24/P21/Q4(b)

4(b)(i)	<table border="1"> <thead> <tr> <th></th><th>equation for reaction</th><th>$\Delta H_{\text{formation}}$</th></tr> </thead> <tbody> <tr> <td>Fe₂O₃</td><td>$2\text{Fe (s)} + \frac{3}{2}\text{O}_2\text{(g)} \rightarrow \text{Fe}_2\text{O}_3\text{(s)}$</td><td>-824.2</td></tr> <tr> <td>CO</td><td>$\text{C (s)} + \frac{1}{2}\text{O}_2\text{(g)} \rightarrow \text{CO (g)}$</td><td>-110.5</td></tr> <tr> <td>CO₂</td><td>$\text{C (s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$</td><td>-393.5</td></tr> </tbody> </table>		equation for reaction	$\Delta H_{\text{formation}}$	Fe ₂ O ₃	$2\text{Fe (s)} + \frac{3}{2}\text{O}_2\text{(g)} \rightarrow \text{Fe}_2\text{O}_3\text{(s)}$	-824.2	CO	$\text{C (s)} + \frac{1}{2}\text{O}_2\text{(g)} \rightarrow \text{CO (g)}$	-110.5	CO ₂	$\text{C (s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$	-393.5	2
	equation for reaction	$\Delta H_{\text{formation}}$												
Fe ₂ O ₃	$2\text{Fe (s)} + \frac{3}{2}\text{O}_2\text{(g)} \rightarrow \text{Fe}_2\text{O}_3\text{(s)}$	-824.2												
CO	$\text{C (s)} + \frac{1}{2}\text{O}_2\text{(g)} \rightarrow \text{CO (g)}$	-110.5												
CO ₂	$\text{C (s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$	-393.5												
4(b)(ii)	$\Delta H_{\text{reaction}} = -\Delta H_{\text{Fe}_2\text{O}_3} - (3 \times -\Delta H_{\text{CO}}) + (3 \times -\Delta H_{\text{CO}_2})$ $= -(-824.2) - (3 \times -110.5) + (3 \times -393.5)$ $= -24.8 \text{ kJ mol}^{-1}$ - evidence of use of correct expression and / or any rearrangement of only these bond energy values - use correct stoichiometry for these bond energy values only - calculate their expression correctly provided the expression uses only addition / subtraction and all 3 values	2												

10. M/J 24/P22/Q2(a)

2(a)	<i>All three correct (2 marks) or two correct (1 mark):</i> - profile for endothermic reaction (include 'hump') - arrow representing E_a - arrow representing ΔH OR (+) 27.2	2
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11. M/J 24/P22/Q2(b)

2(b)(i)	no more fizzing / effervescence stops	1
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2(b)(ii)	M1 calculate Q from $= 50 \times 4.18 \times (26.2 - 19.6) = 1379.4$ M2 Use M1 to find energy change when 1 mol Na_2CO_3 / 2 mol HCl reacts M1 $\div 0.04 = 34485$ M3 express Q as $\Delta H_{(1)}$ in $\text{kJ mol}^{-1} = -(\text{value for M2}) \div 1000$ $= -34.5 \text{ kJ mol}^{-1}$ Alternative method M1 evidence of use $Q = mc\Delta T = 50 \times 4.18 / 1000 \times (26.2 - 19.6) = 1.3794$ M2 Use M1 to find energy change when 1 mol Na_2CO_3 / 2 mol HCl reacts M1 $\div 0.04 = 34.485$ M3 express Q as $\Delta H_{(1)}$ in $\text{kJ mol}^{-1} = -(\text{value for M2})$ (expressed in kJ) $= -34.5 \text{ kJ mol}^{-1}$	3
2(b)(iii)	Using answer from 2(b)(ii) $(=) 2\Delta H_2 - (\Delta H_1)$ $(=) 2 \times 27.2 - (2b(ii))$ $2 \times 27.2 - (-34.50) = (+) 88.9 \text{ kJ mol}^{-1}$ All three correct (2 marks) or two correct (1 mark): $(27.2) - (-34.5)$ OR $(27.2) - (2b(ii))$ 2×27.2 and $1 \times 2b(ii)$ - calculate correctly their expression Using alternative value of $\Delta H_1 = -38.4 \text{ kJ mol}^{-1}$ $\Delta H_r = (+)92.8 \text{ kJ mol}^{-1} = 2 \text{ marks}$	2

12. M/J 24/P23/Q3(a)

3(a)	$\text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l})$ M1 correct equation only M2 correct state symbols based on this equation	2
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13. M/J 24/P23/Q3(b)

3(b)	$\Delta H = -(-1264) - (2 \times -46) + (2 \times -134) + (3 \times -286) = (+)230 \text{ kJ mol}^{-1}$ M1 correct expression in terms of all 4 enthalpy changes of formation AND correct stoichiometry (as described in given equation) M2 correct calculation based on figures given	2
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14. MAR 24/P22/Q3(c)(i)

3(c)(i)	0 for both ΔH_f of N_2 and H_2	1
	-46 for ΔH_f of NH_3	1

15. O/N 23/P21/Q1(e)

1(e)(i)	enthalpy change when one mole of a compound / substance is formed	1
	from its constituent elements in their standard states	1
1(e)(ii)	$2(-592) = 2(-289) + (496) - 2(\text{P=O})$ OR $(2) \text{P=O} = 2(-289) + (496) + 2(592)$	1
	$E_{\text{P=O}} = \frac{1}{2}(1102) = +551 \text{ (kJ mol}^{-1}\text{)}$	1

16. O/N 23/P22/Q3(b)(i)-(ii)

3(b)(i)	enthalpy change when one mole of a compound / substance is formed	1
	from its constituent elements in their standard states	1
3(b)(ii)	$\Delta H_f = +9 + 3(-1281) - 4(-972) = (+)54 \text{ (kJ mol}^{-1}\text{)} (1)$ Any two from (1): - use correct stoichiometry for all three values - use correct expression regardless of stoichiometry for all three values - calculated correctly	2

17. O/N 23/P23/Q1(e)

1(e)(i)	enthalpy change when one mole of a compound / substance is formed	1
	from its constituent elements in their standard states	1
1(e)(ii)	$2(-592) = 2(-289) + (496) - 2(P=O)$ OR $(2) P=O = 2(-289) + (496) + 2(592)$	1
	$E_{P=O} = \frac{1}{2}(1102) = +551 \text{ (kJ mol}^{-1}\text{)}$	1

18. M/J 23/P21/Q2(a)

2(a)	M1 (enthalpy change) when 1 mole of water is formed M2 from an (aqueous) acid and an alkali / (aqueous) base	2
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19. M/J 23/P21/Q2(b)(ii)

2(b)(ii)	M1 correct volume and correct temperature change M2 calculation to calculate Q based on M1 volume and temp and $Q = mc\Delta T$ $Q = mc\Delta T = 175 \times 4.18 \times (27.8 - 20) = (-)5705.7$ M3 energy change when 0.1 mol H_2O is made with appropriate unit and sign $\Delta H = M2 / 0.1 = -57.1 \text{ kJ mol}^{-1}$ OR $-57100 \text{ J mol}^{-1}$	3
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20. M/J 23/P21/Q2(c)(ii)

2(c)(ii)	precipitation (of $BaSO_4$) (also) involves an energy change	1
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21. M/J 23/P22/Q4(a)

4(a)	Method 1 - identification of number and type of bonds when all are broken $2O-H + 4C-H + C=C$ OR $(2 \times 460) + (4 \times 410) + 610$ - Identification of number and type of bonds when all are made $O-H + 5C-H + C-C + C-O$ OR $(460 + (5 \times 410) + 350 + 360)$ - calculation of energy to break bonds – energy to make bonds $3170 - 3220 = -50 \text{ (kJ mol}^{-1}\text{)}$ Two correct for one mark, three correct for two marks. OR Method 2 - identification of number and type of relevant bonds broken (including the new C-H only) $2O-H + C=C$ OR $(2 \times 460) + 610$ - identification of number and type of relevant bonds made (including the new C-H only) $O-H + C-H + C-C + C-O$ OR $(460 + 410 + 350 + 360)$ - calculation using energy to break bonds – energy to make bonds $1530 - 1580 = -50 \text{ (kJ mol}^{-1}\text{)}$ Two correct for one mark, three correct for two marks.	2
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22. M/J 23/P23/Q3(a)

3(a)	M1 (energy change) when 1 mole of water is formed M2 from an (aqueous) acid and an alkali / aqueous base	2
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23. M/J 23/P23/Q3(b)(ii)

3(b)(ii)	M1 correct value for Q (using 0.1 mol H_2O / OR $2 \times (b)(i)$) (and Q in Joules) M2 calculate ΔT (using their Q AND 110) $Q/mc = \Delta T = (Q) \div (4.18 \times 110) = 12.42 \text{ (}^\circ\text{C)}$ M3 use ΔT and initial temperature to find temperature of final solution $21.4 + M2 = 33.8 \text{ (}^\circ\text{C)}$ [M3 $\Delta T + 21.4 = T_{\text{max}}$]	3
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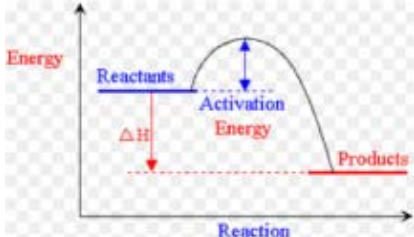
24. M/J 23/P23/Q3(c)(ii)

3(c)(ii)	(additional) energy is involved / required to fully ionise / dissociate (weak / ethanoic acid) OR (additional) energy is involved / required to donate/ release H^+ ions	1
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25. MAR 23/P22/Q3(d)(iii)

3(d)(iii)	$-187 - (-111) - (+52)$	1
	$= -128 \text{ (kJ mol}^{-1}\text{)}$	1

26. M/J 22/P21/Q1(f)(ii)

1(f)(ii)	M1 sketch shows exothermic reaction with a 'hump' AND labelled reactants ($Mg + O_2$) and products (MgO) M2 arrow from reactants / $Mg + O_2$ to products / MgO shown as ΔH M3 arrow showing activation energy / E_a / (+)148 	3
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27. M/J 22/P22/Q1(b)(iii)-(iv)

1(b)(iii)	M1 (enthalpy change / energy change) when one mole of a compound / substance is formed M2 from its elements in their standard states	2
1(b)(iv)	M1 correct method to find amount Mg_3N_2 formed in reaction $3.645 \div 24.3 \text{ mol Mg produces } 0.15 \div 3 \text{ mol of } Mg_3N_2$ OR $0.05 \text{ mol of } Mg_3N_2$ M2 use M1 in correct expression to find energy released by 1 mol Mg_3N_2 $23.05 \div \text{M1}$ OR $23.05 \div \text{mol } Mg_3N_2$ OR $23\,050 \div \text{M1}$ OR $23\,050 \div Mg_3N_2$ M3 calculate value from M2 expression correctly $\Delta H = -(M2) = -461(.0) \text{ kJ mol}^{-1}$ OR $-461\,000 \text{ J mol}^{-1}$	3

28. M/J 22/P22/Q3(b)

3(b)	evaluation, to give a value, based on calculation using: - all three correct bond energies [$HI = 299$, $H_2 = 436$, $I_2 = 151$] - correct use of stoichiometry to calculate ΔH for 1 mol HI $x = 1$ and $y = \frac{1}{2}$ - calculate a value for ΔH correctly i.e. $x(299) - y(436 + 151)$ $\Delta H = (+)5.5 \text{ kJ mol}^{-1}$	2
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29. M/J 22/P23/Q3(a)(ii)

3(a)(ii)	M1 expression OR value derived from using all bond energy values and ΔH correctly, ignoring stoichiometry $-7 = (n350 + m360 + p410) - (q410 + r350 + sC\equiv O)$. M2 correct calculation based on correct expression AND stoichiometry $[-7 = 2(350) + 2(360) + 10(410) - 10(410) - 350 - x] = (+)1077 \text{ kJ mol}^{-1}$	2
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30. MAR 22/P22/Q2(c)(iii)

2(c)(iii)	$-1640 \text{ (kJ mol}^{-1}\text{)}$	1
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